

# Hwaran Lee

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| CONTACT INFORMATION           | Vision AI Labs at T3K, SK Telecom<br>SK T-tower, Eulji-ro 65, Jung-gu, Seoul, South Korea, 04539                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <a href="mailto:hwaran.lee@gmail.com">hwaran.lee@gmail.com</a><br><a href="https://github.com/hwaranlee">hwaranlee.github.io</a> |
| RESEARCH INTERESTS            | Natural Language Processing/Understanding, Dialog Systems, Speech Recognition Systems, Generative Models, Neural Networks and Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                  |
| EDUCATION                     | <b>KAIST</b><br>Ph.D., Electrical Engineering<br>Dissertation: <i>Neural Representations for Speech Recognition and Natural Language Generation</i><br>Advisor: Prof. Soo-Young Lee<br>Committee Members: Prof. Dae-Shik Kim, Hoirin Kim, Alice Oh, Jinwoo Shin, Sang-Hoon Oh<br><br><b>KAIST</b><br>B.S., Mathematical Science<br>Minor: Financial Engineering Program<br><i>Magna Cum Laude</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Daejeon, South Korea<br>Mar. 2013 – Aug. 2018<br><br>Daejeon, South Korea<br>Feb. 2008 – Aug. 2012                               |
| RESEARCH AND WORK EXPERIENCES | <b>Research Scientist</b> <ul style="list-style-type: none"><li>T-Brain, AI Center, SK Telecom, Seoul, South Korea</li></ul> <b>Graduate Research Assistant</b> <ul style="list-style-type: none"><li>Electrical Engineering, KAIST, Daejeon, South Korea</li></ul> <b>Undergraduate Researcher</b> <ul style="list-style-type: none"><li>Brain Science Research Center, Daejeon, South Korea</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Nov. 2018 – Present<br><br>Mar. 2013 – Aug. 2018<br><br>Jun. 2012 – Feb. 2013                                                    |
| PUBLICATIONS                  | <b>International Journal</b><br><br>[J1] <b>Hwaran Lee</b> , Seokhwan Jo, HyungJun Kim, Sangkeun Jung, and Tae-Yoon Kim, “SUMBT+LaRL: End-to-end Neural Task-oriented Dialog System with Reinforcement Learning”, <i>arXiv preprint arXiv:2009.10447</i> , 2020.<br><br>[J2] Geonmin Kim, <b>Hwaran Lee</b> , Bo-Kyeong Kim, Sang-Hoon Oh, and Soo-Young Lee, “Unpaired Speech Enhancement by Acoustic and Adversarial Supervision for Speech Recognition”, <i>IEEE Signal Processing Letters</i> , (2019): 159-163.<br><br>[J3] Ho-Gyeong Kim, <b>Hwaran Lee</b> , Geonmin Kim, Sang-Hoon Oh, and Soo-Young Lee, “Rescoring of N-best Hypotheses using Top-down Selective Attention for Automatic Speech Recognition”, <i>IEEE Signal Processing Letters</i> , (2018): 199-203.<br><br>[J4] <b>Hwaran Lee</b> , Geonmin Kim, Ho-Gyeong Kim, Sang-Hoon Oh, and Soo-Young Lee, “Deep CNNs Along the Time Axis With Intermap Pooling for Robustness to Spectral Variations”, <i>IEEE Signal Processing Letters</i> 23.10 (2016): 1310-1314.<br><br>[J5] <b>Hwaran Lee</b> , Nadeem Iqbal, Wonil Chang, and Soo-Young Lee, “A Calibration Method for Eye-Gaze Estimation Systems Based on 3D Geometrical Optics”, <i>IEEE Sensors Journal</i> 13, no. 9 (2013): 3219-3225.<br><br>[J6] Nadeem Iqbal, <b>Hwaran Lee</b> , and Soo-Young Lee, “Smart User Interface for Mobile Consumer Devices Using Model-Based Eye-Gaze Estimation”, <i>IEEE Transactions on Consumer Electronics</i> 59, no. 1 (2013): 161-166. |                                                                                                                                  |

## International Conference

- [C1] Gi-Cheon Kang, Junseok Park, **Hwaran Lee**, Byoung-Tak Zhang, and Jin-Hwa Kim, “DialGraph: Sparse Graph Learning Networks for Visual Dialog”, arXiv preprint arXiv:2004.06698, 2020.
- [C2] **Hwaran Lee**<sup>\*</sup>, Jinsik Lee<sup>\*</sup>, and Tae-Yoon Kim, “SUMBT: Slot-Utterance Matching for Universal and Scalable Belief Tracker”, *The 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.
- [C3] Geonmin Kim, **Hwaran Lee**, Bo-Kyeong Kim, and Soo-Young Lee, “Compositional Sentence Representation from Character within Large Context Text”, *International Conference on Neural Information Processing (ICONIP)*, 2017.
- [C4] Ho-Gyeong Kim, Jihyeon Roh, **Hwaran Lee**, Geonmin Kim, and Soo-Young Lee, “Active Learning for Large-scale Object Classification: from Exploration to Exploitation” *In Proceedings of the 3rd International Conference on Human-Agent Interaction, (HAI)*, 2015.
- [C5] Bo-Kyeong Kim, **Hwaran Lee**, Jihyeon Roh, and Soo-Young Lee, “Hierarchical committee of deep CNNs with exponentially-weighted decision fusion for static facial expression recognition”, *In Proceedings of the 2015 ACM on International Conference on Multimodal Interaction (ICMI)*, 2015.

## Conference without Proceedings

- [W1] Geonmin Kim<sup>\*</sup>, **Hwaran Lee**<sup>\*</sup>, CheongAn Lee, Eunmi Hong, Byunggeun Kim, Soo-Young Lee, “A Deep Chatbot for QA and Chitchat.”, The Conversational Intelligence Challenge section on NIPS 2017 Competition Track Workshop, 2017.
- [W2] **Hwaran Lee**, Geonmin Kim, Jihyeon Roh, and Soo-Young Lee, “Learning Tonotopically Organized Auditory Feature-map from Speech by an Intermap Pooling Layer in a Deep CNN”, *15th China-Japan-Korea Joint Workshop on Neurobiology and Neuroinformatics (NBNI)*, 2015. (only abstract)
- [W3] Geonmin Kim, **Hwaran Lee**, Jaemyung Yu, and Soo-Young Lee, “Spoken Sentence Embedding from Character by Jointly Learning Character-level Compositional Word Model and RNN Sentence Encoder”, *15th China-Japan-Korea Joint Workshop on Neurobiology and Neuroinformatics (NBNI)*, 2015. (only abstract)

## PATENTS

- [P1] Tae-Yoon Kim, Jin Kim, Hyungjoon Kim, Jinsik Lee, **Hwaran Lee**, Heewon Jeon, Seokhwan Jo, “Method and Apparatus for Providing Hybrid Intelligent Customer Consultation”,
  - Korea Patent Application 10-2019-0136035, filed October 2019, Patent Pending.
  - PCT/KR2020/008832, filed July 2020, Patent Pending.
- [P2] **Hwaran Lee**, Jinsik Lee, Tae-Yoon Kim, “Method and Apparatus for Dialogue State Tracking for Use in Goal-oriented Dialog System”, Korea Patent Application 10-2019-0086380, filed July 2019, Patent Pending.

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<sup>\*</sup>these authors contributed equally to this work

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| RESEARCH<br>PROJECTS     | <b>Meta Learner Project</b>                                                                                                                                                                                                                                                                                                                                                                                                     | 2020 - Present        |
|                          | <ul style="list-style-type: none"> <li>Funded by SK Telecom <ul style="list-style-type: none"> <li>Researched and developed semi-supervised learning algorithms and modules for AutoTrainer</li> </ul> </li> </ul>                                                                                                                                                                                                              |                       |
|                          | <b>End-to-end Goal-Oriented Dialog Systems</b>                                                                                                                                                                                                                                                                                                                                                                                  | 2018 - 2019           |
|                          | <ul style="list-style-type: none"> <li>Funded by SK Telecom <ul style="list-style-type: none"> <li>Researched and developed a scalable multi-domain natural language understanding</li> <li>Researched and developed end-to-end neural goal-oriented dialog systems</li> <li>Participated in the End-to-End Multi-Domain Task-Completion Task (DSTC8 Track1 Task1 at AAAI 2020) and ranked 6th place</li> </ul> </li> </ul>     |                       |
|                          | <b>A Deep Chatbot for QA and Chitchat</b>                                                                                                                                                                                                                                                                                                                                                                                       | 2017                  |
|                          | <ul style="list-style-type: none"> <li>Funded by <i>Institute for Information &amp; Communications Technology Promotion (IITP)</i> <ul style="list-style-type: none"> <li>Researched and developed an article-based chatbot that carry on both question-answering and chitchat conversations</li> <li><b>Ranked 3rd Place</b> in the Conversational Intelligence Challenge of NIPS 2017 Live Competition</li> </ul> </li> </ul> |                       |
|                          | <b>Deep Learning Based Korean Understanding and Applications</b>                                                                                                                                                                                                                                                                                                                                                                | 2015 – 2018           |
|                          | <ul style="list-style-type: none"> <li>Funded by HANCOM Inc. <ul style="list-style-type: none"> <li>Researched and developed Korean morpheme, syllable-level language models, sentence and document models</li> <li>Researched and developed continual learning algorithms for language models</li> </ul> </li> </ul>                                                                                                           |                       |
|                          | <b>Speech Feature Extraction Based on Deep Learning for Continuous Speech Recognition Systems</b>                                                                                                                                                                                                                                                                                                                               | 2013 – 2014           |
|                          | <ul style="list-style-type: none"> <li>Funded by LG Electronics <ul style="list-style-type: none"> <li>Developed convolutional neural networks for acoustic models of continuous English speech</li> <li>Programmed C/C++ and CUDA based on KALDI toolkit</li> </ul> </li> </ul>                                                                                                                                                |                       |
|                          | <b>Development of Dialog-based Spontaneous Speech Interface Technology on Mobile Platforms</b>                                                                                                                                                                                                                                                                                                                                  | 2013 – 2014           |
|                          | <ul style="list-style-type: none"> <li>Funded by Electronics and Telecommunications Research Institute (<i>ETRI</i>) <ul style="list-style-type: none"> <li>Developed deep belief networks and auto-encoders for acoustic models of Korean syllable data</li> <li>Programmed models in C/C++ and CUDA languages</li> </ul> </li> </ul>                                                                                          |                       |
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| HONORS AND<br>AWARDS     | <ul style="list-style-type: none"> <li>Annual Roll Award, KAIST EE</li> <li>Ranked 3rd, ConvAI challenge, NIPS 2017 Competition Track Workshop</li> </ul>                                                                                                                                                                                                                                                                       | Apr. 2018             |
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2017                  |
|                          | <ul style="list-style-type: none"> <li>Challenge Winner, ICMI EmotiW2015</li> </ul>                                                                                                                                                                                                                                                                                                                                             | 2015                  |
|                          | <ul style="list-style-type: none"> <li>Best Paper Award, HAI</li> </ul>                                                                                                                                                                                                                                                                                                                                                         | 2015                  |
|                          | <ul style="list-style-type: none"> <li>Qualcomm Innovation Award</li> </ul>                                                                                                                                                                                                                                                                                                                                                     | 2015                  |
|                          | <ul style="list-style-type: none"> <li>BK21 Plus Financial Support for Graduates Long Term Training</li> </ul>                                                                                                                                                                                                                                                                                                                  | May. 2014             |
|                          | <ul style="list-style-type: none"> <li>KAIST Graduate Scholarship</li> </ul>                                                                                                                                                                                                                                                                                                                                                    | Mar. 2013 - Aug. 2018 |
|                          | <ul style="list-style-type: none"> <li>Australian Endeavour Student Exchange Grant (AUD\$ 5000),<br/>The University of Queensland</li> </ul>                                                                                                                                                                                                                                                                                    | Apr. 2011             |
|                          | <ul style="list-style-type: none"> <li>National Excellence Scholarship, KOSAF</li> </ul>                                                                                                                                                                                                                                                                                                                                        | Feb.2008 - Feb. 2012  |
|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                       |
| ACADEMIC PEER<br>REVIEWS | <ul style="list-style-type: none"> <li>The International Conference on Learning Representations (<i>ICLR</i>)</li> <li>International Conference on Computational Linguistics (<i>COLING</i>)</li> </ul>                                                                                                                                                                                                                         | 2021<br>2020          |

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|                        | <ul style="list-style-type: none"> <li>• Samsung Humantech Paper Awards Committee 2020</li> <li>• Qualcomm Innovation Awards (Undergraduate Track) Committee 2019</li> <li>• Speech Communication 2019</li> <li>• IEEE Transactions on Neural Networks and Learning Systems 2017 - 2018</li> <li>• Neural Processing Letters 2015</li> </ul>                                                                                                                                              |
| INVITED TALKS          | <ul style="list-style-type: none"> <li>• Introduction to deep learning for dialogue systems <ul style="list-style-type: none"> <li>Inha Univ., Seoul, Oct. 2020</li> <li>Yeonsei Univ., Seoul, Jun. 2020</li> <li>Sookmyung Women's Univ., Seoul, Oct. 2019</li> </ul> </li> <li>• Toward end-to-end neural dialog systems for multi-domain task completion <ul style="list-style-type: none"> <li>KAIST, Daejeon, Dec. 2019</li> </ul> </li> </ul>                                       |
| TEACHING<br>EXPERIENCE | <p><b>Teaching Assistant</b></p> <ul style="list-style-type: none"> <li>• ChatBot PyTorch Implementation, KB-KAIST AI Sep. 2017</li> <li>• EE476 Audio-Visual Perception Models Spring 2016, 2017</li> <li>• EE538 Neural Networks Fall 2015, 2016, 2017</li> </ul> <p><b>Graduate Mentor</b></p> <ul style="list-style-type: none"> <li>• Undergraduate Mentoring Program, KAIST Counselling Center <ul style="list-style-type: none"> <li>Fall 2014, Spring 2016</li> </ul> </li> </ul> |
| SKILLS                 | <ul style="list-style-type: none"> <li>• <b>Languages</b> Python, Lua, MATLAB, C/C++, CUDA</li> <li>• <b>Libraries</b> PyTorch, TensorFlow, Torch7, KALDI</li> </ul>                                                                                                                                                                                                                                                                                                                      |
| REFERENCES             | <p>Prof. Soo-Young Lee (Ph.D. advisor)</p> <p>Emeritus Professor, School of Electrical Engineering, KAIST      <a href="mailto:sylee@kaist.ac.kr">sylee@kaist.ac.kr</a></p> <p>Director, KAIST Institute for Artificial Intelligence      <a href="http://cnsl.kaist.ac.kr">http://cnsl.kaist.ac.kr</a></p>                                                                                                                                                                               |